

1)

a)

$$I_{DS(\text{lin})} = \frac{W}{L} \mu_n C_{ox} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

We are going to do a linear extrapolation and ignore the $\frac{V_{DS}^2}{2}$ term

$$I_{DS(\text{lin})} = 13 \mu\text{A} \text{ when } V_{GS} = 0.55 \text{V}$$

$$I_{DS(\text{lin})} = 27.5 \mu\text{A} \text{ when } V_{GS} = 0.65 \text{V}$$

Using plot 1

$$\frac{I_{DS(0.55\text{V})}}{I_{DS(0.65\text{V})}} = \frac{\frac{W}{L} \mu_n C_{ox} [(0.55 - V_T) V_{DS}]}{\frac{W}{L} \mu_n C_{ox} [(0.65 - V_T) V_{DS}]}$$

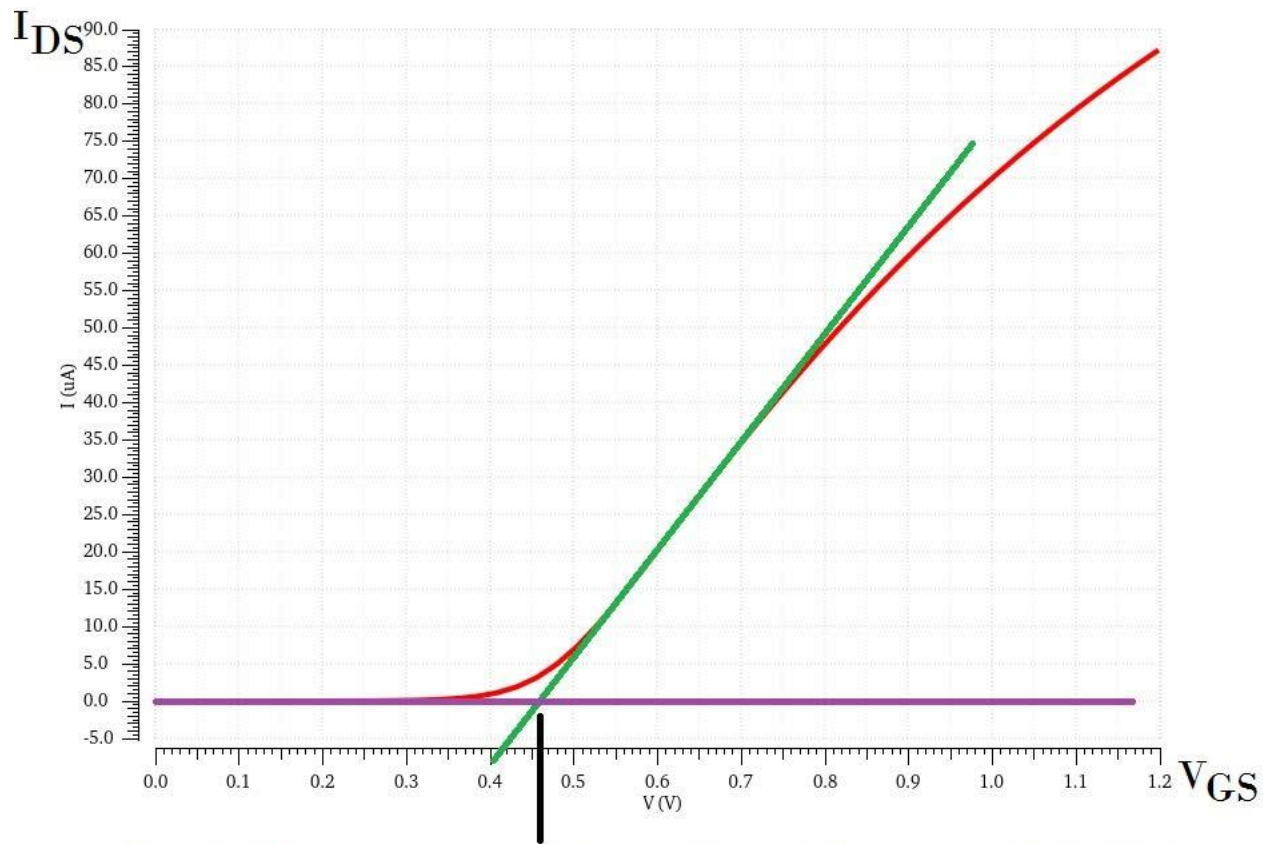
$V_{DS} = 0.1 \text{V}$, other terms cancel

$$\frac{13}{27.5} = \frac{(0.55 - V_T) V_{DS}}{(0.65 - V_T) V_{DS}} = \frac{0.55 - 0.1 V_T}{0.65 - 0.1 V_T}$$

$$\frac{13}{27.5} = \frac{0.55 - V_T}{0.65 - V_T}$$

$$\boxed{V_T = 0.46 \text{V}}$$

Inspection of plot method to obtain V_{T0} from plot 1:



V_{T0} is defined as the intersection point of V_{GS} and $I_{DS(lin)}$ where $I_{DS(lin)} = 0$ when $V_{SB} = 0$
Here this intersection is 0.46V for $V_{T0} = 0.46V$

b) Plot 2, $I_{DS}(V_{GS}=1.2V, V_{DS}=1.2V) = 265 \mu A$

$I_{DS}(V_{GS}=1.2V, V_{DS}=0.8V) = 248 \mu A$

Using $V_T = 0.46V$

$V_{DS} > V_{GS} - V_T$

$0.8 > 0.74$

saturated (borely)

$$\frac{I_{DS}(V_{DS}=1.2V)}{I_{DS}(V_{DS}=0.8V)} = \frac{\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS2})}{\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS1})}$$

$$\frac{265}{248} = \frac{1 + \lambda(1.2)}{1 + \lambda(0.8)}$$

$$\frac{265}{248} = \frac{1 + \lambda(1.2)}{1 + \lambda(0.8)}$$

$$\boxed{\lambda = 0.2}$$

c] Plot 3

$$I_{DS}(V_{SB}=0) = 265 \mu A$$

$$I_{DS}(V_{SB}=0.4) = 233 \mu A$$

$$V_{DS} > V_{GS} - V_T$$

Assume V_T is close to V_{T0}

$$1.2 > 0.74$$

saturated

$$\frac{I_{DS}(V_{SB}=0)}{I_{DS}(V_{SB}=0.4)} = \frac{\frac{1}{2} \mu_n C_{ox} (V_{GS} - V_{T0})^2}{\frac{1}{2} \mu_n C_{ox} (V_{GS} - V_{T1})^2}$$

$$\frac{265}{233} = \frac{(1.2 - 0.46)^2}{(1.2 - V_{T1})^2}$$

$$V_{T1} = 0.506 V$$

$$V_{T1} = V_{T0} + \gamma (\sqrt{V_{SB} + 2\phi_F} - \sqrt{2\phi_F})$$

$$0.506 = 0.46 + \gamma (\sqrt{0.4 + 0.88} - \sqrt{0.88})$$

$$\boxed{\gamma = 0.24 V^{1/2}}$$

d) For μ_n , solve for μ_n at any I_{DS}

$$I_{DS}(V_{GS}=1.2, V_{DS}=1.2, V_{SB}=0) = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

$$265 \mu A = \frac{1}{2} \mu_n (1.1 \cdot 10^{-6}) \cdot 5 \cdot (1.2 - 0.46)^2 (1 + 0.2 \cdot 1.2)$$

$$\boxed{\mu_n = 142 \text{ cm}^2/V_s}$$

If $\mu_n = 350 \text{ cm}^2/V_s$,

$$I_{DS} = \frac{1}{2} 350 \cdot (1.1 \cdot 10^{-6}) \cdot 5 \cdot (1.2 - 0.46)^2 \cdot (1 + 0.2 \cdot 1.2)$$

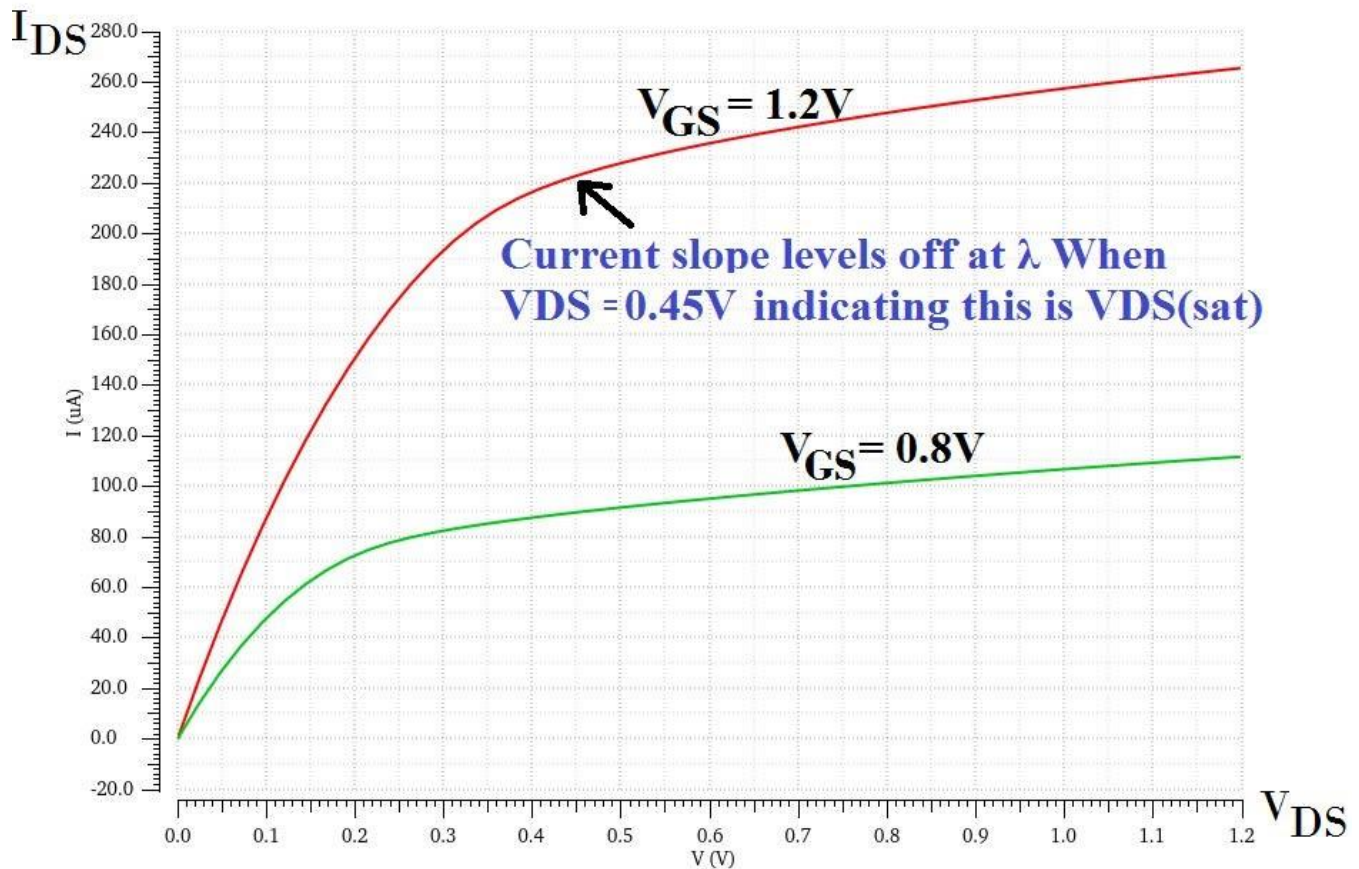
$$\boxed{I_{DS} = 654 \mu A}$$

Achieving a low μ_n of $142 \text{ cm}^2/V_s$, and an I_{DS} of $654 \mu A$ versus the expected $265 \mu A$ when a realistic μ_n was used, indicates that the I_{DS} formula we are using does not adequately describe the NMOS. Short channel effects are likely not being accounted for.

e]

The onset of saturation is when $V_{DS} = V_{GS} - V_T$
 When $V_{GS} - V_T = 1.2 - 0.46$, the expectation is

$$V_{DS(sat)} = 0.74 \text{ V}$$



The observed $V_{DS(sat)}$ is 0.29V lower than expected.

f)

From part d, $V_D(\text{sat}) = 0.5V$

$$V_D(\text{sat}) = \frac{(V_{GS} - V_T)(E_c L)}{V_{GS} - V_T + E_c L}$$

$$V_{GS} - V_T + E_c L$$

$$0.45 = \frac{(1.2 - 0.46) E_c L}{1.2 - 0.46 + E_c L}$$

$$E_c L = 1.15V$$

$$I_{DS}(\text{sat}) = W V_{\text{sat}} C_{\text{ox}} \frac{(V_{GS} - V_T)^2}{V_{GS} - V_T + E_c L} (1 + \lambda V_{DS})$$

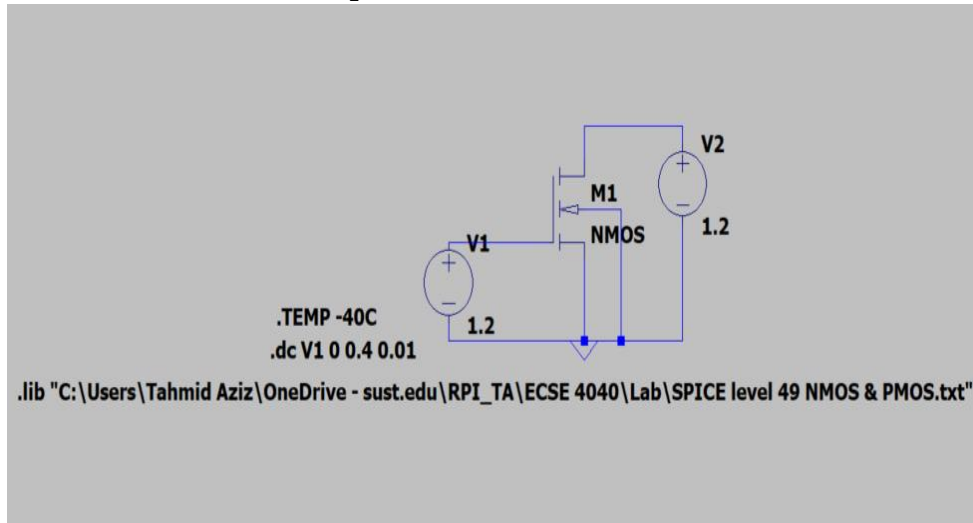
$$I_{DS}(\text{sat}) = (600 \cdot 10^{-7}) \cdot 10^7 \cdot (1.1 \cdot 10^{-6}) \cdot \frac{(1.2 - 0.46)^2}{1.2 - 0.46 + 1.15} (1 + 0.2 \cdot 1.2)$$

$$I_{DS}(\text{sat}) = 237 \mu A$$

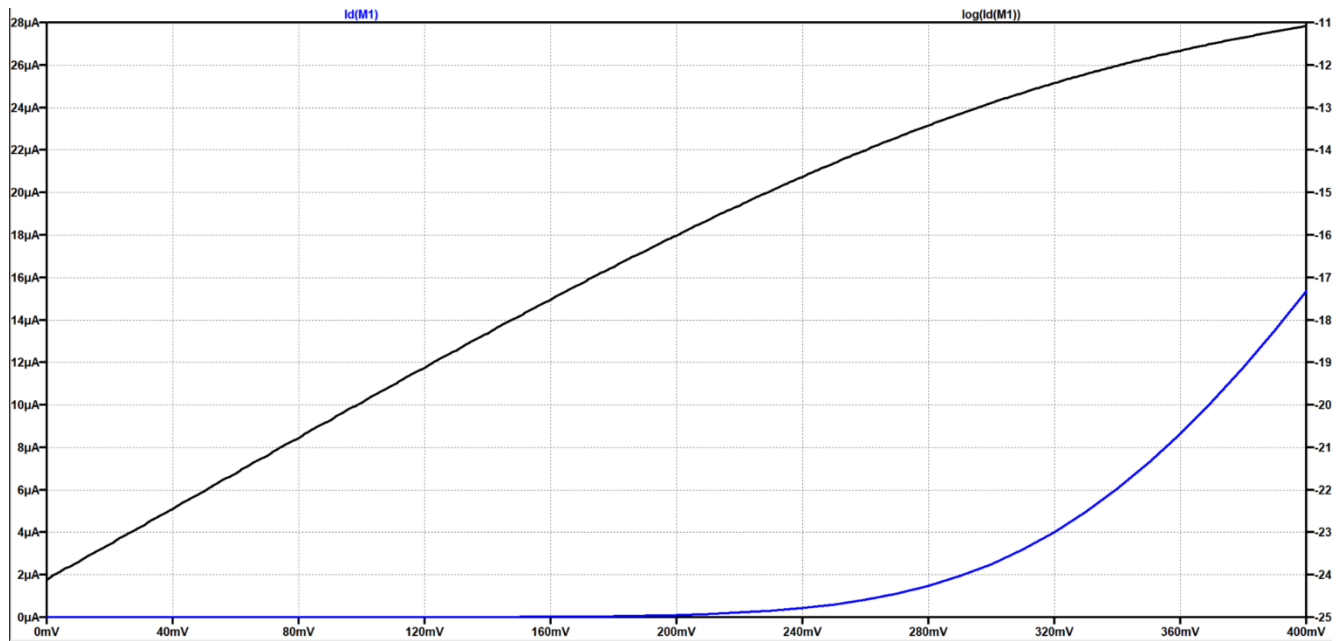
The $237 \mu A$ result where critical field limitation is considered is much closer to the expected $265 \mu A$ result than the λ effect only result of $654 \mu A$ in part d.

Question 2

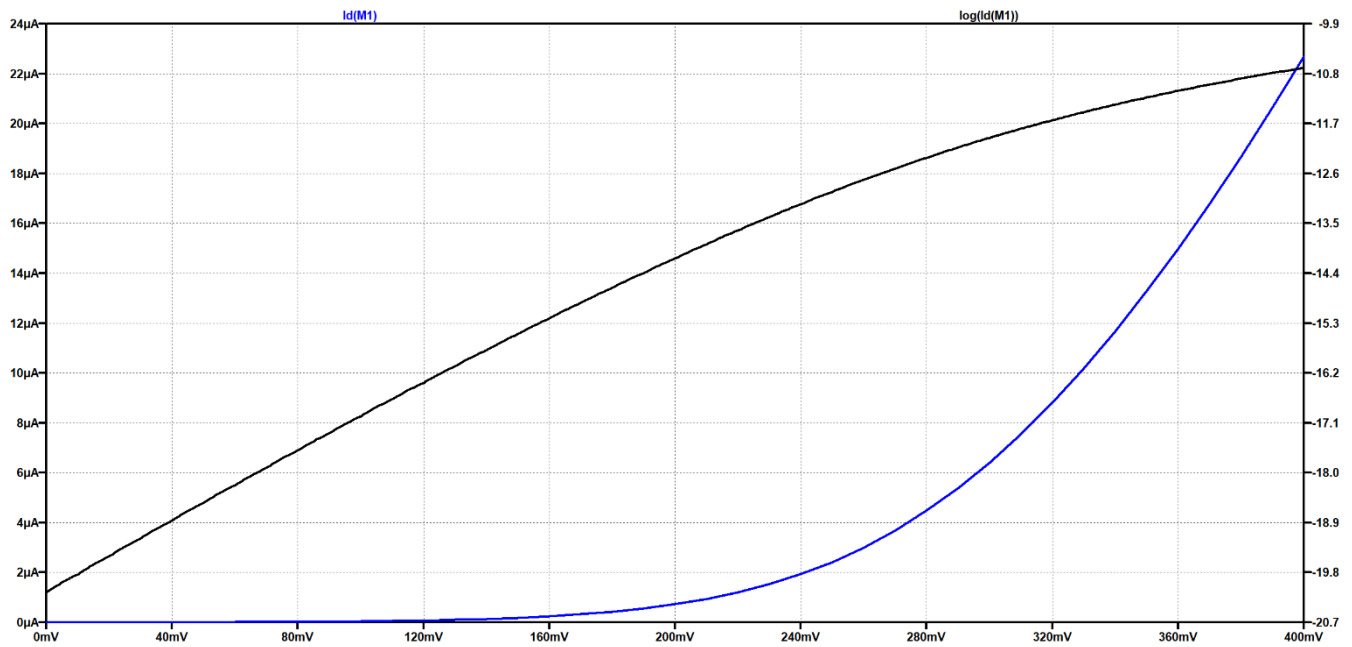
LTspice schematic



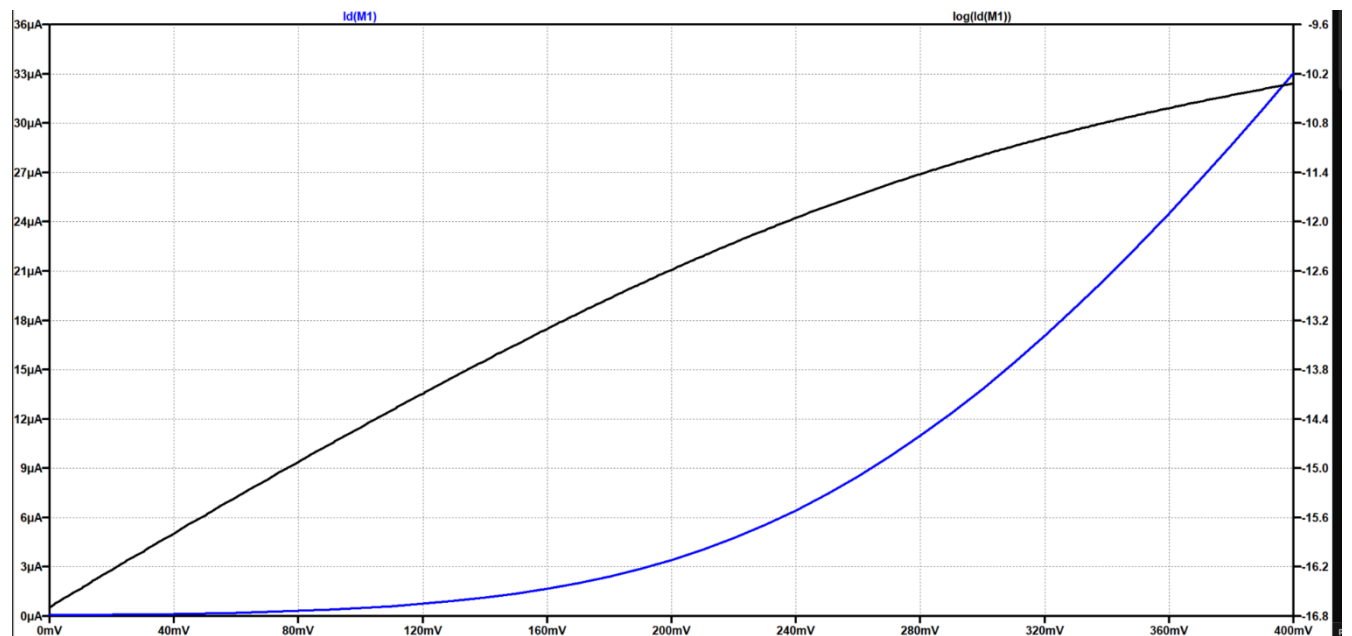
Subthreshold conduction plots



-40C, peak I_{ds} 15 μA



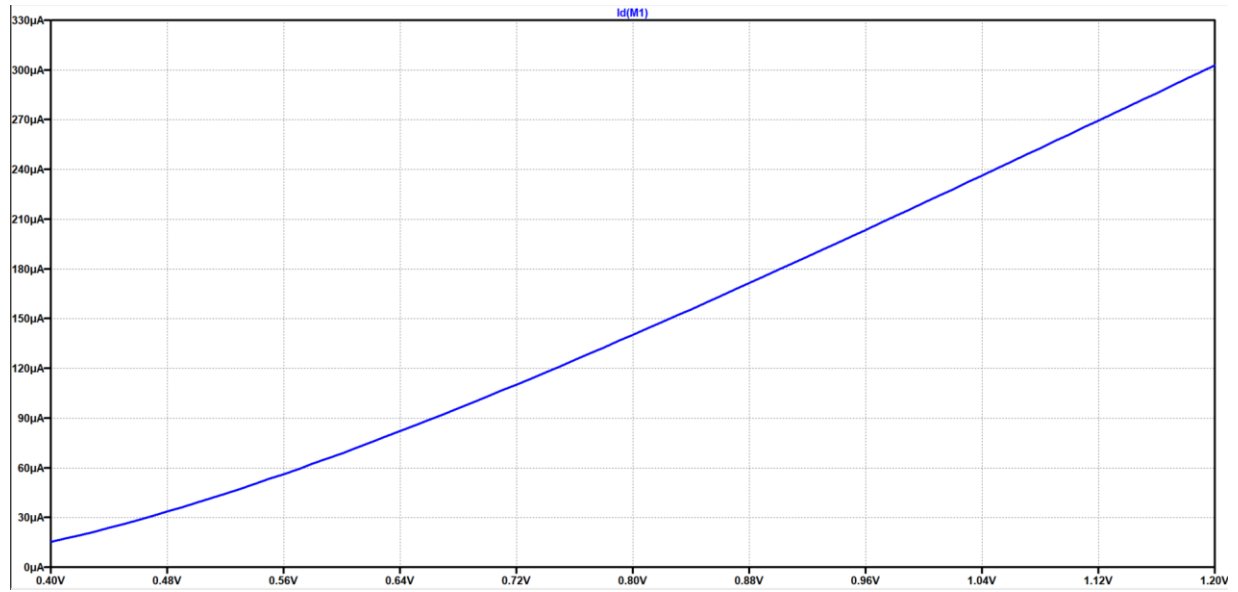
25C, peak I_{ds} 23 μA



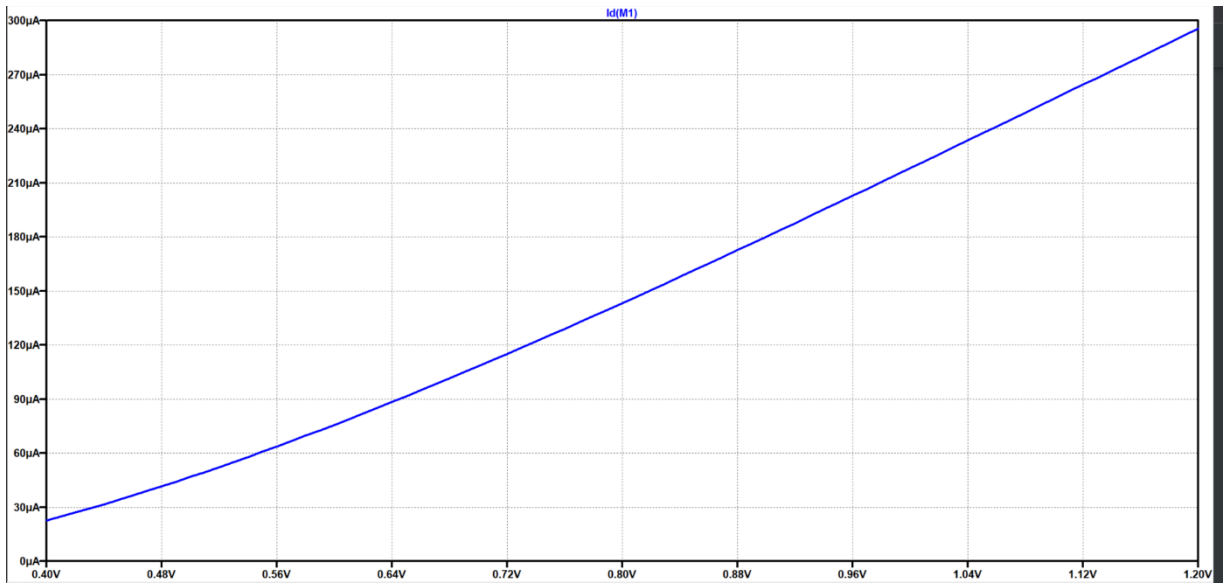
125C, peak I_{ds} 33 μA

In the subthreshold region we want our transistor to be off and conduct zero current. The -40C scenario conducts the least current and 125C scenario the most. Therefore, lower temperatures results in more ideal behavior.

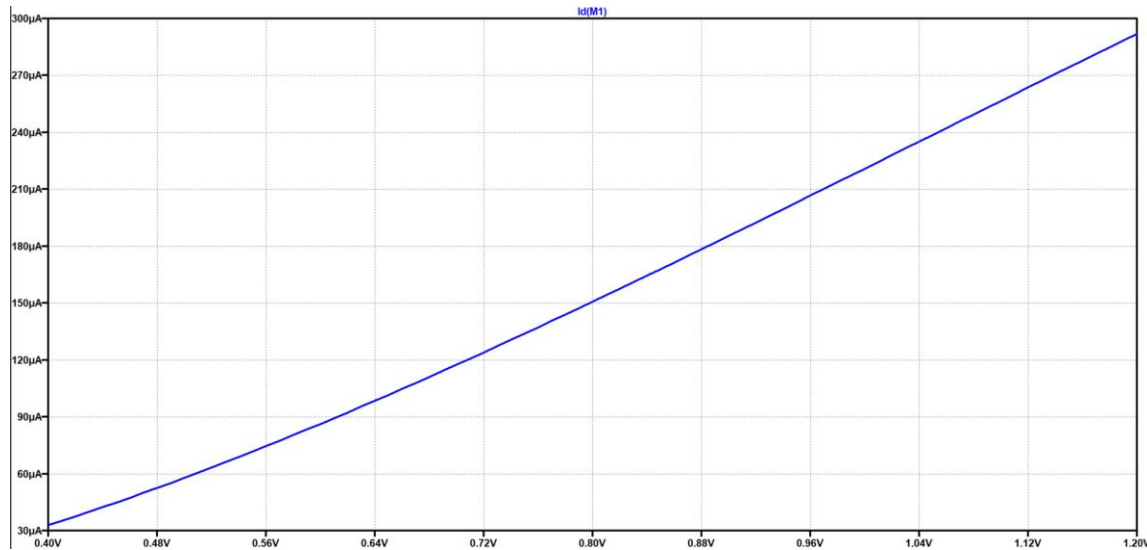
Active region plots



-40C, peak I_{ds} 303 μA



25C, peak I_{ds} 295 μA



125C, peak Ids 290 μ A

When the transistor is in the active region (linear or saturated), it is desirable to pass as much current as possible. -60C passed the most current and 125C the least, therefore, lower temperatures are desired for improved device behavior.